

Chapter 30

Using the Inspector in the Cut Page

The Inspector holds all the controls to modify, resize, retime, and generally adjust anything related to a clip, transition, or effect on the Cut page Timeline.

Contents

Using the Inspector.....	598
Video.....	598
Audio.....	606
Effects.....	611
Transition.....	612
Image.....	613
File.....	614

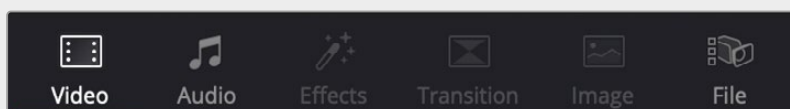
Using the Inspector

The Inspector has been redesigned to make it easier to find specific controls and to adjust common settings for your clips. Instead of a long vertical list, different aspects of the Inspector have now been organized into panels, with each controlling specific grouped sets of parameters for your clip.

The Inspector is activated by clicking on the Inspector Panel in the upper-right section of the User Interface toolbar. The Inspector is broken up into individual Video, Audio, Effects, Transition, Image, and File panels. Inspector panels that are not applicable to your clip or selection are grayed out.



The Inspector Panel icon in the upper right of the UI toolbar



The Inspector panels showing Video, Audio, and File parameters available for adjustment; others are grayed out.

Methods of using controls in the Inspector:

- **To activate or deactivate a control:** Click the toggle to the left of the control's name. The orange dot on the right means the control is activated. A gray dot on the left means the control is deactivated.
- **To reveal a control's parameters:** Double-click the control's name.
- **To reset controls to their defaults:** Click the reset button to the right of the control's name.

Video

The Video panel of the Inspector exposes a vast array of controls designed to manipulate the size, speed, and opacity of your clips.

Transform



The Transform section of the Video Inspector panel

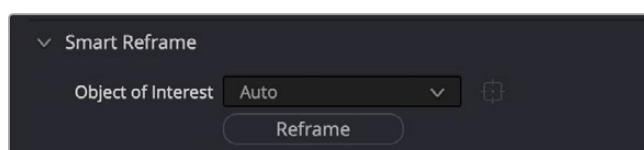
The Transform group includes the following parameters for resizing and repositioning your clips:

- **Zoom X and Y:** Allows you to blow the image up or shrink it down. The X and Y parameters can be linked to lock the aspect ratio of the image, or released to stretch or squeeze the image in one direction only.
- **Position X and Y:** Moves the image within the frame, allowing pan and scan adjustments to be made. X moves the image left or right, and Y moves the image up or down.
- **Rotation Angle:** Rotates the image around the anchor point.
- **Anchor Point X and Y:** Defines the coordinate on that clip about which all transforms are centered.
- **Pitch:** Rotates the image toward or away from the camera along an axis running through the center of the image, from left to right. Positive values push the top of the image away and bring the bottom of the image forward. Negative values bring the top of the image forward and push the bottom of the image away. Higher values stretch the image more extremely.
- **Yaw:** Rotates the image toward or away from the camera along an axis running through the center of the image from top to bottom. Positive values bring the left of the image forward and push the right of the image away. Negative values push the left of the image away and push the right of the image forward. Higher values stretch the image more extremely.
- **Flip Image:** Two buttons let you flip the image in different dimensions.

Flip Horizontal control: Reverses the image along the X-axis, left to right.

Flip Vertical control: Reverses the clip along the Y-axis, turning it upside down.

Smart Reframe (Studio Version Only)



The Smart Reframe section of the Video Inspector panel

The Smart Reframe feature makes it easier to quickly reframe material across extreme aspect ratio changes. It's useful for situations where you've shot a 16:9 horizontal video and find yourself needing to create a vertically-oriented 9:16 version for mobile phones and social media deliverables, or using 4:3 archival footage in a 2.39:1 widescreen movie. Smart Reframe can be used manually, or automatically executed using the DaVinci Resolve Neural Engine.

- **Object of Interest:** Tools for selecting the subject that the resize will frame around.
 - Auto:** DaVinci Resolve's Neural Engine will analyze the clip and choose its most representative object. This will be the only option if more than one clip is selected for Smart Reframing.
 - Reference Point:** Allows you to manually adjust a bounding box around the subject to reframe around.
- **Reframe:** This button executes the Smart Reframe command. This can take some time depending on the length and number of clips.

Cropping

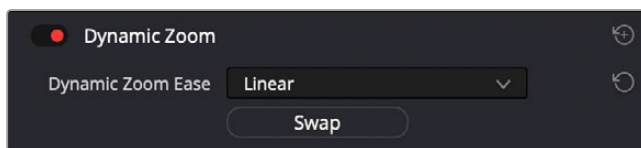


The Cropping section of the Video Inspector panel

The Video Inspector controls the image's cropping parameters.

- **Crop Left, Right, Top, and Bottom:** Lets you cut off, in pixels, the four sides of the image. Cropping a clip creates transparency so that whatever is underneath shows through.
- **Softness:** Lets you blur the edges of a crop. Setting this to a negative value softens the edges inside of the crop box, while setting this to a positive value softens the edges outside of the crop box.
- **Retain Image Position:** Clicking this checkbox will lock the crop parameters in place when you resize the image using the Transform tool above. Unchecking this box will scale and position the crop as well as the image.

Dynamic Zoom



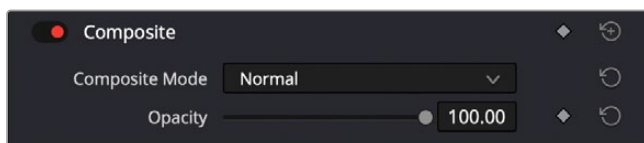
The Dynamic Zoom section of the Video Inspector panel

The Dynamic Zoom controls, which are off by default, make it fast and easy to do pan and scan effects to zoom into or out of a clip. Turning the Dynamic Zoom group on activates two controls in the Inspector that work hand-in-hand with the Dynamic Zoom onscreen adjustment controls.

For more information on using the Dynamic Zoom controls, see *Chapter 50, “Compositing and Transforms in the Timeline.”*

- **Dynamic Zoom Ease:** Lets you choose how the motion created by these controls accelerates. You can choose from Linear, Ease In, Ease Out, and Ease In and Out.
- **Swap:** This button reverses the start and end transforms that create the dynamic zoom effect.

Composite



The Composite section of the Video Inspector panel

Composite modes can be used to combine clips that are superimposed over other clips in the Timeline.

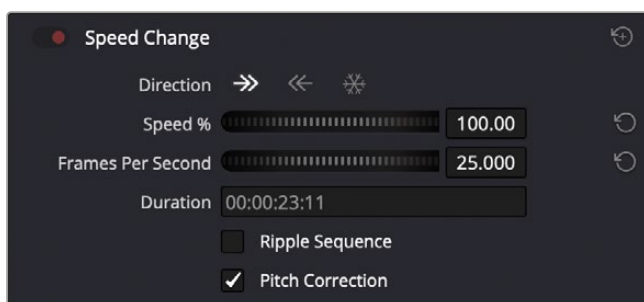
- **Composite Mode:** This selects the type of composite mode to combine the superimposed clips. The default “Normal” means no compositing mode is applied.

For more information on Composite Modes, see *Chapter 50, “Compositing and Transforms in the Timeline.”*

- **Opacity:** This slider makes a clip more or less transparent in addition to compositing already being done.

Speed Change

You can change the speed of your clip directly in the Video Inspector’s Speed Change controls. This method has the benefit of being available in both Cut and Edit pages.



The Speed Change controls in the Video Inspector

- **Direction:** Selects the desired motion of the clip, forward, backward, or freeze frame.
- **Speed %:** Adjusting this slider changes the clips motion on a percentage basis. This value can be keyframed.
- **Frames Per Second:** Adjusting this slider changes the clips motion by increasing or decreasing the number of frames per second to play the clip back at. This value can be keyframed.
- **Duration:** You can directly select how long you want the clip to be by setting a specific duration here in HH:MM:SS:FF format. This will then automatically adjust the speed of the clip to playback all frames in that exact amount of time.
- **Ripple Sequence checkbox:** If you want the speed change you’re about to make to ripple the Timeline, pushing or pulling all clips following the current one to accommodate the clip’s new size, then turn on this checkbox.
- **Pitch Correction checkbox:** Checking this box will perform pitch correction on the audio attached to the clip so that while the audio duration is changed to match the picture speed, it will still sound natural. Be aware that pitch correction on large speed adjustments may not sound as good as pitch corrections made to small speed adjustments.

Stabilization



The Stabilization section of the Video Inspector panel

These controls let you smooth out or even steady unwanted camera motion within a clip.

The analysis is performed in such a way as to preserve the motion of individual subjects within the frame, as well as the overall direction of desirable camera motion, while correcting for unsteadiness.

These are the same stabilizer controls found in the Color page's Tracker palette (minus the tracker graph), and the resulting stabilization analysis is mirrored on the Color page, where you can see the data visualized on the graph, if necessary.

A pop-up menu provides three different options that determine how the selected clip is analyzed and transformed during stabilization. You must choose an option first, before clicking the Stabilize button above, because the option you choose changes how the image analysis is performed. If you choose another option, you must click the Stabilize button again to reanalyze the clip.

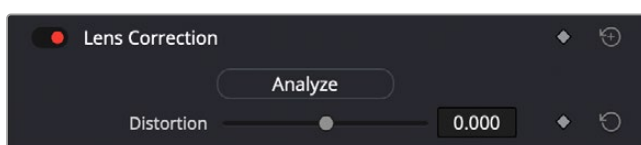
- **Perspective:** Enables perspective, pan, tilt, zoom, and rotation analysis and stabilization.
- **Similarity:** Enables pan, tilt, zoom, and rotation analysis and stabilization, for instances where perspective analysis results in unwanted motion artifacts.
- **Translation:** Enables pan and tilt analysis and stabilization only, for instances where only X and Y stabilization gives you acceptable results.

The other controls let you customize how aggressively the selected clip is stabilized.

- **Stabilization Toggle:** The toggle control for the Stabilization controls lets you turn stabilization off and on to be able to compare the stabilized and unstabilized image.
- **Camera Lock:** Turning on this checkbox disables Cropping Ratio and Smooth, and enables the stabilizer to focus on eliminating all camera motion from the shot in an effort to create a locked shot.
- **Zoom:** When this checkbox is turned on, the image is resized by a large enough percentage to eliminate the blanking (black edges) that is the result of warping and transforming the image to eliminate unwanted camera motion. The lower a value Cropping Ratio is set to, the more DaVinci Resolve will need to zoom into an image to eliminate these blanked edges. If you turn this off, the image is not zoomed at all, and whatever blanking intrudes into the image is output along with the image, on the assumption that you'll have dedicated compositing artists deal with eliminating this blanking by filling in the missing image data in a more sophisticated manner. You may also leave this checkbox turned off if you're planning on animating the Input Sizing Zoom parameter to dynamically zoom into and out of a shot being stabilized to eliminate blanking only where it occurs, using only as much zooming as is necessary for each region of the shot.
- **Cropping Ratio:** This value limits how hard the stabilizer tries to stabilize, by dictating how much blanking or zooming you're willing to accept in exchange for eliminating unwanted motion. A value of 1.0 results in no stabilization being applied. Progressively lower values enable more aggressive stabilization. Changing this value requires you to click the Stabilize button again to reanalyze the clip.

- **Smooth:** Lets you apply mathematical smoothing to the analyzed data used to stabilize the clip, allowing camera motion in the shot while eliminating unwanted jittering. Lower values perform less smoothing, allowing more of the character of the original camera motion to show through, while higher values smooth the shot more aggressively. Changing this value requires you to click the Stabilize button again to reanalyze the clip.
- **Strength:** This value is a multiplier that lets you choose how tightly you want to use the stabilization track to eliminate motion from a shot using the current analysis. With a value of 1, stabilization is maximized. Since some clips might look more natural with looser stabilization, choosing a number lower than 1 lets a percentage of the original camera motion show through. Zero (0) disables stabilization altogether. As an additional tip, you can invert the stabilization by choosing -1 when pasting a stabilization analysis from another clip to perform a match move based on the overall motion of the scene, and you can use a negative value either lower than 0 or higher than -1 to under or overcompensate when inverting the stabilization, simulating the effects of parallax where foreground and background planes move together but at different speeds.

Lens Correction

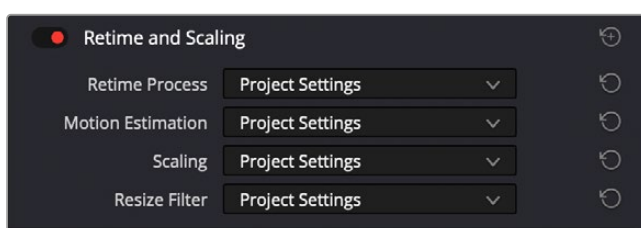


The Lens Correction section of the Video Inspector panel

The Lens Correction group (only available in Resolve Studio) has two controls that let you correct for lens distortion in the image, or add lens distortion of your own.

- **Analyze:** Automatically analyzes the frame in the Timeline at the position of the playhead for edges that are being distorted by wide angle lens. Clicking the Analyze button moves the Distortion slider to provide an automatic correction. If you're analyzing a particularly challenging clip, a progress bar will appear to let you know how long this will take.
- **Distortion:** Dragging this slider to the right lets you manually apply a warp to the image that lets you straighten the bent areas of the picture that can be caused by wide angle lenses. If you clicked the Analyze button and the result was an overcorrection, then dragging this slider to the left lets you back off of the automatic adjustment until the image looks correct.

Retime and Scaling



The Retime and Scaling section of the Video Inspector panel

The Retime and Scaling group has four parameters that affect retiming quality and clip scale:

- **Retime Process:** Lets you choose a default method of processing clips in mixed frame rate timelines and those with speed effects (fast forward or slow motion) applied to them, on a clip-by-clip basis. The default setting is "Project Settings," so all speed-effected clips are treated the same way.

There are three options: Nearest, Frame Blend, and Optical Flow, which are explained in more detail in the Speed Effect Processing section see *Chapter 51, “Speed Effects.”*

- **Motion estimation mode:** When using Optical Flow to process speed change effects or clips with a different frame rate than that of the Timeline, the Motion Estimation pop-up lets you choose the best-looking rendering option for a particular clip. Each method has different artifacts, and the highest quality option isn't always the best choice for a particular clip. The default setting is “Project Settings,” so all speed-effected clips are treated the same way. There are several options. The “Standard Faster” and “Standard Better” settings are the same options that have been available in previous versions of DaVinci Resolve. They're more processor efficient and yield good quality that are suitable for most situations. However, “Enhanced Faster” and “Enhanced Better” should yield superior results in nearly every case where the standard options exhibit artifacts, at the expense of being more computationally intensive, and thus slower on most systems.
 - The “Standard Faster” and “Standard Better” settings are the same options that have been available in previous versions of DaVinci Resolve. They're more processor efficient and yield good quality that are suitable for most situations. However, “Enhanced Faster” and “Enhanced Better” should yield superior results in nearly every case where the standard options exhibit artifacts, at the expense of being more computationally intensive, and thus slower on most systems.
 - “Speed Warp Faster and “Speed Warp Better” are available for even higher-quality slow motion effects using the DaVinci Neural Engine. Your results with this setting will vary according to the content of the clip, but in ideal circumstances this will yield higher visual quality with fewer artifacts than even the Enhanced Better setting.
- **Scaling:** Lets you choose how clips that don't match the current project resolution are handled on a clip-by-clip basis. The default setting is “Project Settings,” so that all mismatched clips use the same method of being automatically resized. However, you can also choose an individual method of automatic scaling for any clip.

The options are Crop, Fit, Fill, and Stretch; for more information see *Chapter 152, “Sizing and Image Stabilization.”*

- **Resize Filter:** For clips that are being resized in any way, this setting lets you choose the filter method used to interpolate image pixels when resizing clips. Different settings work better for different kinds of resizing. There are four options:
 - Sharper:** Usually provides the best quality in projects using clips that must be scaled up to fill a larger frame size, or scaled down to HD resolutions.
 - Smoother:** May provide higher quality for projects using clips that must be scaled down to fit an SD resolution frame size.
 - Bicubic:** While the Sharper and Smoother options are slightly higher quality, Bicubic is still an exceptionally good resizing filter and is less processor intensive than either of those options.
 - Bilinear:** A lower quality setting that is less processor intensive. Useful for previewing your work on a low-performance computer before rendering, when you can switch to one of the higher quality options.
 - Other Resize Methods:** A selection of specific resize algorithms is available if you need to match them to other VFX workflows.

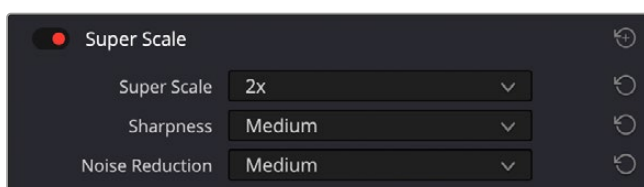
- **Deinterlace Quality (Interlaced Clips Only):** Allows per clip deinterlace quality adjustments regarding how DaVinci Resolve combines the two fields of interlaced media into progressive frames.

Normal: A high-quality deinterlacing method that is suitable for most clips. For many clips, Normal is indistinguishable from High. Normal is always used automatically during playback in DaVinci Resolve.

High: A more processor-intensive method that can sometimes yield better results, depending on the footage, at the expense of slower rendering times.

DaVinci Neural Engine: This option uses the advanced machine learning algorithms of the DaVinci Neural Engine to analyze motion between the fields of interlaced material and reconstructs them into a single frame. This option is very computationally intensive but, ideally, will deliver an even more aesthetically pleasing result than the “high” setting.

Super Scale



The Super Scale parameters

For instances when you need higher-quality upscaling than the standard Resize Filters allow, you can now enable one of the “Super Scale” options in the Inspector. Unlike using one of the numerous scaling options in the Edit, Fusion, or Color pages, Super Scale actually increases the source resolution of the clip being processed, which means that clip will have more pixels than it did before and will be more processor-intensive to work with than before, unless you optimize the clip (which bakes in the Super Scale effect into the optimized media) or cache the clip in some way.

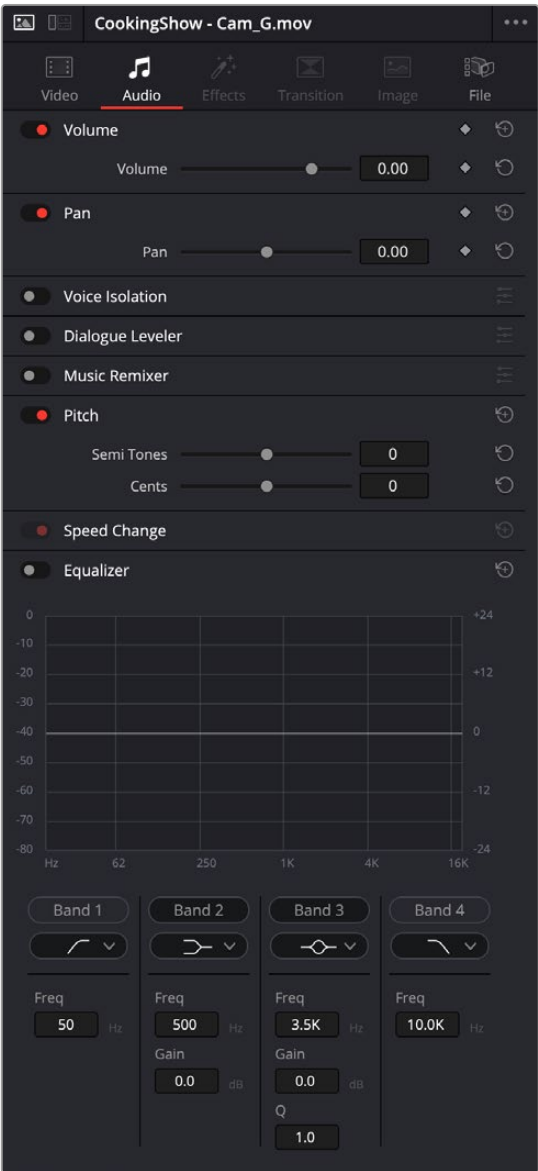
For more detailed information on Super Scale, see *Chapter 11, “Image Sizing and Resolution Independence.”*

Using Super Scale in the Inspector is functionally equivalent to setting the same controls for the media clip in the Clip Attributes. This means that changing this setting affects all of the additional edits referencing the selected media as well.

The Super Scale group has the following parameters that affect the quality and clip scale:

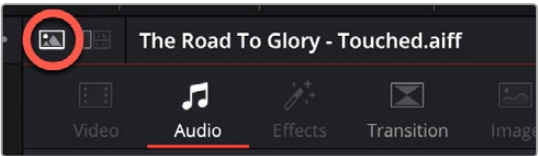
- **Super Scale:** Lets you choose the amount of scaling required. The options are 2-3-4x and 2-3-4x Enhanced.
- **Sharpness:** Lets you choose the amount of detail in the scaling. This is limited to Low, Medium, or High, unless the Super Scale mode is set to Enhanced, which allows you to apply variable sharpness. You will want to balance this setting against Noise Reduction.
- **Noise Reduction:** Lets you choose the amount of noise reduction in the scaling. This is limited to Low, Medium, or High, unless the Super Scale mode is set to Enhanced, which allows you to apply variable noise reduction. You will want to balance this setting against Sharpness.

Audio

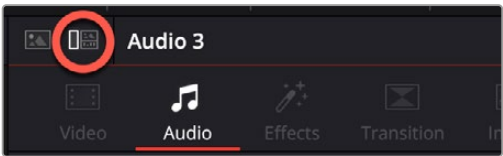


The Audio Inspector Clip and Track parameters

The Audio Inspector can work on both individual clips or entire tracks, based on which icon is selected in the upper left of the Inspector. Normally this option automatically adjusts for you based on what you have selected in the timeline. For example, when you click on an audio clip in the timeline, the Inspector will open in Clip mode, and if you click on an audio track header in the timeline, the Inspector will switch to Track mode automatically. These can always be manually overridden simply by clicking on the appropriate icon. Clip and Track modes contain different options for audio editing.



The Audio Inspector in Clip mode (circled)



The Audio Inspector in Track mode (circled)

The Audio tab contains commonly used audio controls for video editing purposes, including Volume, Pan, Pitch, and Equalizer. It also contains several incredibly useful DaVinci Neural Engine powered tools like Voice Isolation, Dialog Leveler, and Music Remixer.

- **Volume:** Each clip has a single volume control that corresponds to the volume overlay over each audio clip.
- **Pan:** A simple Pan slider that controls stereo panning. While most professional mixes will restrict panning to the more robust panner found in the Fairlight page Mixer, this simple Pan control is useful for editors of visuals working in the Edit page to quickly create simple panning effects to aid in a craft edit.
- **Voice Isolation (Studio Version Only):** Voice Isolation is a plugin that can remove loud, undesirable sounds from existing voice recordings. This effect utilizes AI to provide models that let you completely remove undesired sounds. The AI model is trained for any type of human voice, male or female, young or old, so you can get incredible results that isolate dialog from background sounds in a recording, including everything from air conditioning or fans to extremely loud sounds like a jackhammer, restaurant background noise, or music playing at the same time that the subject is speaking, and so on.

For more information about Voice Isolation, see *Chapter 31, "Video and Audio Effects in the Cut Page."*

Amount: Lets you adjust the amount of voice isolation processing.

For more information about Voice Isolation, see *Chapter 181, "Fairlight FX."*

- **Dialogue Leveler:** The Dialogue Leveler analyzes source material to detect dialog and then "ride down" louder areas, "lift up" softer areas, and lower background sounds that are not dialog. It works without typical dynamics processor "pumping" (compression/limiting) or other obvious side effects and produces results similar to detailed manual clip gain adjustments or by carefully "riding" the track with fader automation.

Preset Menu: The preset menu provides the following options:

- **Optimized for most sources:** This option is selected by default and works well for most sources.
- **More lift for low levels:** Select this option if the source has more low level dialog that you want to boost.
- **Lift soft whispery sources:** Select this option if the source has whispered dialog and background noise.

Reduce Loud Dialogue: When enabled, louder dialogue is ridden downward on peaks and acts somewhat like a "perfect limiter" where you don't have to adjust threshold or time constants. Due to the "near real time" aspect, analysis occurs prior to audible playback for optimal results.

Lift Soft Dialogue: When enabled, finds low level dialogue and lifts and evens out material that is more variable in level and softer, but because the process is dialogue focused, it doesn't tend to raise background sounds (unless they are happening at the same time as the dialogue itself). More often than not, the Lift Soft Dialogue option is the most useful of the three options, as it makes less audible lines of dialogue more audible and naturally smooth, while not boosting background noise.

Background Reduction: When enabled, reduces background sounds by focusing on dialogue and gently removing them based on the internal presets (Preset menu).

Output Gain: Adjust the Output Gain by clicking and dragging on the Output Gain control or by entering a value in the numeric field (Output Gain control in dB with 0 to +6 dB range with .1dB resolution).

- **Dialogue Separator (Track Mode Only):** The Dialogue Separator uses DaVinci Neural Engine AI to give you individual control over the level of dialogue, background sounds and “ambience,” the reverberant field, or ambient room sound. If you have a source file that contains a great roomy sound, but there is music in the background or crowd noises, you can rebalance so that the background sounds come down a bit, or adjust the room sound down a bit for better intelligibility.

For more information about the Dialogue Separator, see *Chapter 31, “Video and Audio Effects in the Cut Page.”*

- **Music Remixer:** The Music Remixer is a DaVinci Neural Engine AI-based effect. This Track FX plugin uses AI to split music into individual basic stems: Vocals, Drums, Bass, Guitar and “Other,” which means “everything else.” You can use the level controls to rebalance or remix the sources, or the mute buttons to make instant changes and bring parts in and out of your mix. Music Remixer lets you creatively adjust a music track to suit your needs. For example, you might have a track where you need to remove the vocal or rearrange to allow for quieter moments or a build.

For more information about the Music Remixer, see *Chapter 31, “Video and Audio Effects in the Cut Page.”*

- **Music Editor:** Allows you to seamlessly extend or shorten music to the length of your edit. You can enter a desired duration and press the Adjust button in order to shorten or extend the clip to that duration. Music Editor does not use time compression/expansion or alter the pitch of the music, but instead uses editing techniques like a music editor might to fit the material to desired length to picture. It will look for logical places to create transitions based on whether you’re extending length (by repeating sections), or removing material, and attempt to use the ending or fade of the original composition, if it times well. If the ending is abrupt, it’s up to you to manually create a fade out.

For more information about the Music Editor, see *Chapter 31, “Video and Audio Effects in the Cut Page.”*

NOTE: Music Editor is an “intelligent edit” that works effectively with beat driven music (pop, dance, etc.). It is not intended for free form, ambient, or non-beat driven material.

- **Ducker (Track Mode Only):** The Ducker Track FX plugin allows you to automatically lower the level of a track by using an external source signal – this is known as “ducking.” Most commonly, ducking is used to have dialogue or VO automatically lower a music or sound effects bed, but it can be used in many creative ways. Set up correctly, your sources can often “mix themselves,” and you can create very useful results.

The Ducker uses changes in level only to do its work; the audio is not compressed or limited. For more information about the Ducker, see *Chapter 31, “Video and Audio Effects in the Cut Page.”*

Choose the target track that you want to affect (the “destination” track), and open the Ducker.

Source: This is the source track you want to use to duck the level of the target track. You can add multiple or remove additional source tracks by pressing the +/- buttons to the right.

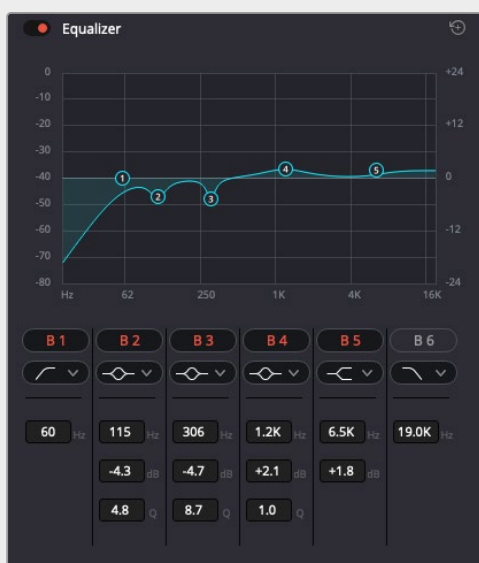
Duck Level: Adjust the amount of “ducking” applied.

- **Pitch:** Lets you alter the pitch of a clip without changing the speed. Two sliders let you adjust clip pitch in semi tones (large adjustments, a twelfth of an octave) and cents (fine adjustments, 100th of a semi tone).

Equalizer

This six-band parametric equalizer has both graphical and numeric controls for boosting or attenuating different ranges of frequencies within that clip, before it even gets to the EQ built into the Mixer. Each band has controls for the filter type (Bell, Lo-Shelf, Hi-Shelf, Notch), Frequency, Gain, and Q-factor (sharpness of the band), with the available controls for each band of EQ changing depending on the filter type.

TIP: Clip EQ settings can be copied and pasted from one Clip EQ to another, and between Clip EQ and Track EQ instances.



Clip EQ in the Audio Inspector

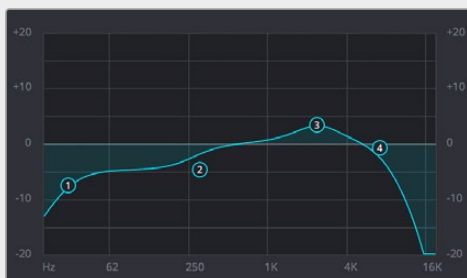
Master EQ Controls

The Equalizer window has the following overall controls located in the upper-right and left corners:

- **Enable button:** Turns the overall EQ effect off and on, without resetting the controls.
- **Reset button:** Resets all controls of the EQ window to their defaults.

Clip EQ Graph

This graph displays the current EQ curve with handles that correspond to each enabled EQ band listed below. You can drag any numbered handle to boost or attenuate the range of frequencies governed by that band, using whatever type of equalization that band has been set to.



The Clip EQ graph with user-draggable handles

Dragging the numbered handles on this graph in turn modifies the parameters of the corresponding band, and changing each band's parameters will also alter the EQ graph, which serves the additional purpose of providing a graphical representation of the equalization being applied to that track.

Bands 1 and 6

The outer two sets of band controls let you make high-pass and low-pass adjustments, if necessary.

- **Band enable button:** Turns each band of EQ on and off.
- **Band filter type:** Bands 1 and 6 can be switched among five specific filtering options for processing the lowest or highest frequencies in the signal.
- **For Band 1, these include (from top to bottom):** Lo-Shelf, Bell, Notch, Hi-Shelf, and Hi-Pass.
- **For Band 6, the options are (from top to bottom):** Lo-Pass, Lo-Shelf, Bell, Notch, and Hi-Shelf.
- **Freq:** Adjusts the center frequency of the EQ adjustment.
- **Gain:** Adjusts the amount by which the affected frequencies are impacted. Negative values attenuate those frequencies, while positive values boost those frequencies.
- **Q Factor:** This parameter adjusts the width of affected frequencies and appears whenever the Bell option is selected. Lower values include a wider range of frequencies; higher values include a narrower range of frequencies.

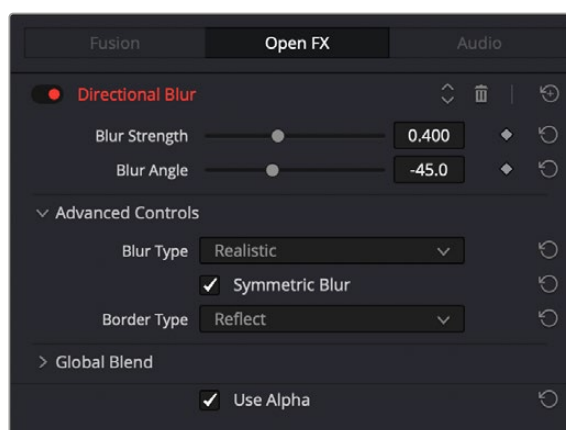
Bands 2 through 5

These band controls let you make a wide variety of equalization adjustments.

- **Band enable button:** Turns each band of EQ on and off.
- **Band filter type:** These Bands offer four filtering options (from top to bottom): Lo-Shelf, Bell, Notch, and Hi-Shelf.
- **Frequency:** Adjusts the center frequency of the EQ adjustment.
- **Gain:** Adjusts the amount by which the affected frequencies are impacted. Negative values attenuate those frequencies, while positive values boost those frequencies.
- **Q Factor:** This parameter appears when the Bell setting is selected. It adjusts the width of affected frequencies. Lower values include a wider range of frequencies; higher values narrow the range of frequencies.

NOTE: There are many more refined plugins and effects for audio clips in the Audio FX library. If you apply any of these, the controls will appear in the Inspector's Effects tab Audio section, instead of here.

Effects



The Effects Inspector controls

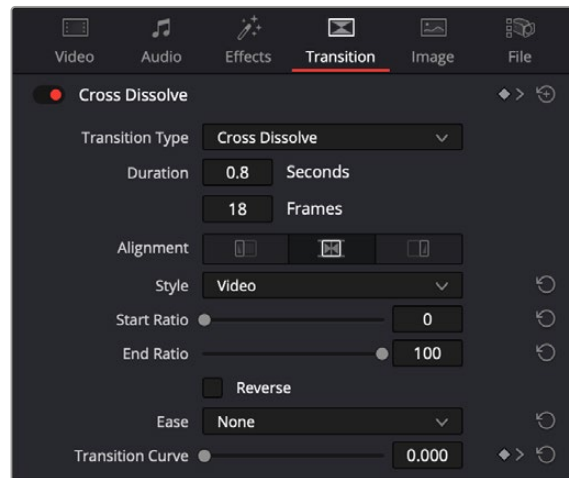
Any Fusion FX, Open FX, or Audio FX filters that have been applied to a clip can be modified here in their respective tabs. Different effects in the Timeline expose different controls in the Effects panel. Whichever panels are exposed, parameters within each panel are organized into groups, with a title bar providing the name of that group, along with other controls that let you control all parameters within that group at the same time.

These controls include:

- **Enable button:** A toggle control to the left of the parameter group's name lets you disable and re-enable every parameter within that group at once. Orange means that track's enabled. Gray is disabled.
- **Parameter group title bar:** Double-clicking the title bar of any group of parameters collapses or opens them. Even more exciting than that, Option-double-clicking the title bar of one parameter group collapses or opens all parameter groups at once.
- **Keyframe and Next/Previous Keyframe buttons:** This button lets you add or remove keyframes at the position of the playhead to or from every single parameter within the group. When the button is highlighted orange, a keyframe is at the current position of the playhead. When it's dark gray, there is no keyframe. Left and right arrow buttons let you jump the playhead from keyframe to keyframe for further adjustment.
- **Reset button:** Lets you reset all parameters within that group to their default settings.
- **Use Alpha:** Checking this box applies the Open FX alpha channel to the selected clip, compositing it over any background elements that appear in lower tracks. If more than one alpha-modifying effect is applied to a single clip, the alpha channels are mixed together.

For a detailed explanation of each of the Resolve FX plugins that accompany DaVinci Resolve, see *Part 12, "Resolve FX Overview."*

Transition



The Transition Inspector controls

Double-clicking a transition in the Timeline opens that Transition Panel in the Inspector. Each transition has the following properties you can edit.

- **Transition Type:** The currently selected transition. You can change to any other installed transition by selecting one in the drop-down menu.
- **Duration:** The duration of the transition, shown in both seconds and frames.
- **Alignment:** A drop-down that lets you choose the transition's position relative to the edit point it's applied to. Your choices are "Start on Edit," "Center on Edit," and "End on Edit."

Additional properties that are specific to each type of transition appear in another group below. Since the Cross Dissolve transition is the most common transition used, its properties will be shown as an example.

- **Style:** The different Dissolve transitions (Cross Dissolve, Additive Dissolve, and so on) expose this drop-down that lets you choose different ways for the outgoing clip to blend into the incoming clip during the dissolve. There are six different options to choose from:

Video: A simple linear dissolve; the outgoing clip fades out as the incoming clip fades in.

Film: A logarithmic dissolve, simulating film dissolves as created by an optical printer.

Additive: The outgoing and incoming clips are cross faded using the Additive composite mode. As a result, the transition seems to brighten at the halfway point.

Subtractive: The outgoing and incoming clips are cross faded using the Subtractive composite mode. As a result, the transition seems to darken at the halfway point.

Highlights: The outgoing and incoming clips are cross faded using the Lighten composite mode. The lightest parts of each clip are emphasized during this transition.

Shadows: The outgoing and incoming clips are cross faded using the Darken composite mode. The darkest parts of each clip are emphasized during this transition.

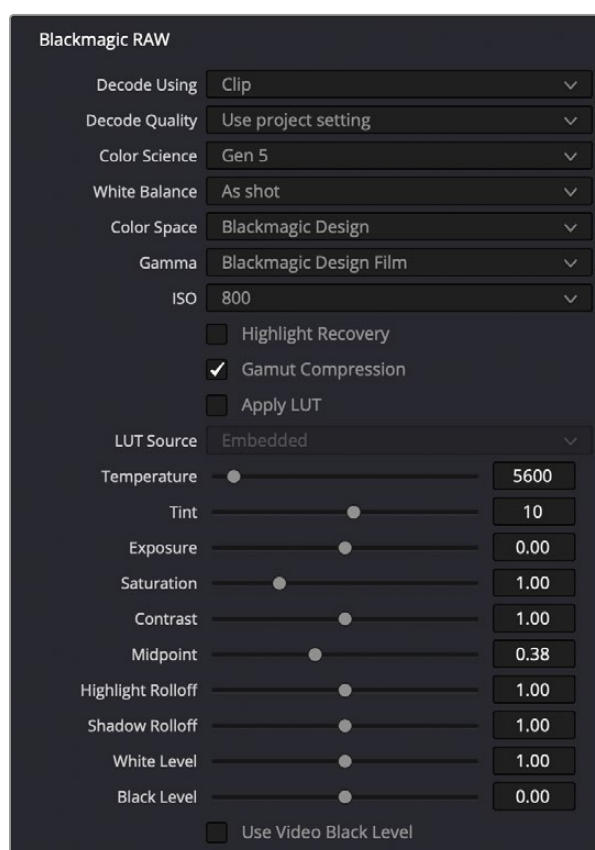
- **Start Ratio:** Defines the percentage of completion for the transition at its first frame, from 0 to 100 percent. Setting the Start Ratio to anything but 0 results in the transition immediately appearing at a more fully cross-dissolved state from the very first frame.
- **End Ratio:** Defines the percentage of completion for the transition at its last frame. Setting the End Ratio to anything but 0 results in the transition never fully dissolving to the incoming shot at its last frame.

- **Reverse:** Reverses the transition. This parameter is disabled for Dissolve transitions.
- **Ease:** A drop-down that lets you apply nonlinear acceleration to the beginning, ending, or overall duration of a transition. The result is to add inertia to the transition from the outgoing clip to the incoming clip, providing a gentler change from each clip into and out of the transition.
 - None:** The outgoing clip fades away to the next shot in a linear fashion.
 - In:** The outgoing clip lingers as the beginning of the transition dissolves more slowly than the end.
 - Out:** The outgoing clip fades away more quickly as the beginning of the transition dissolves more quickly than the end.
 - In & Out:** Both the outgoing and incoming clips make slower transitions at the beginning and end of the dissolve, but the very center of the transition is faster as a result.
 - Custom:** Lets you modify the parameters of the fade manually using the Transition Curves below.
- **Transition Curve:** Allows you to manually set keyframes controlling the progress of the transition along its duration.

Other types of transitions display properties that are specific to that transition's particular effect.

For a detailed explanation of each of the transitions that accompany DaVinci Resolve, see *Chapter 48, "Using Transitions."*

Image

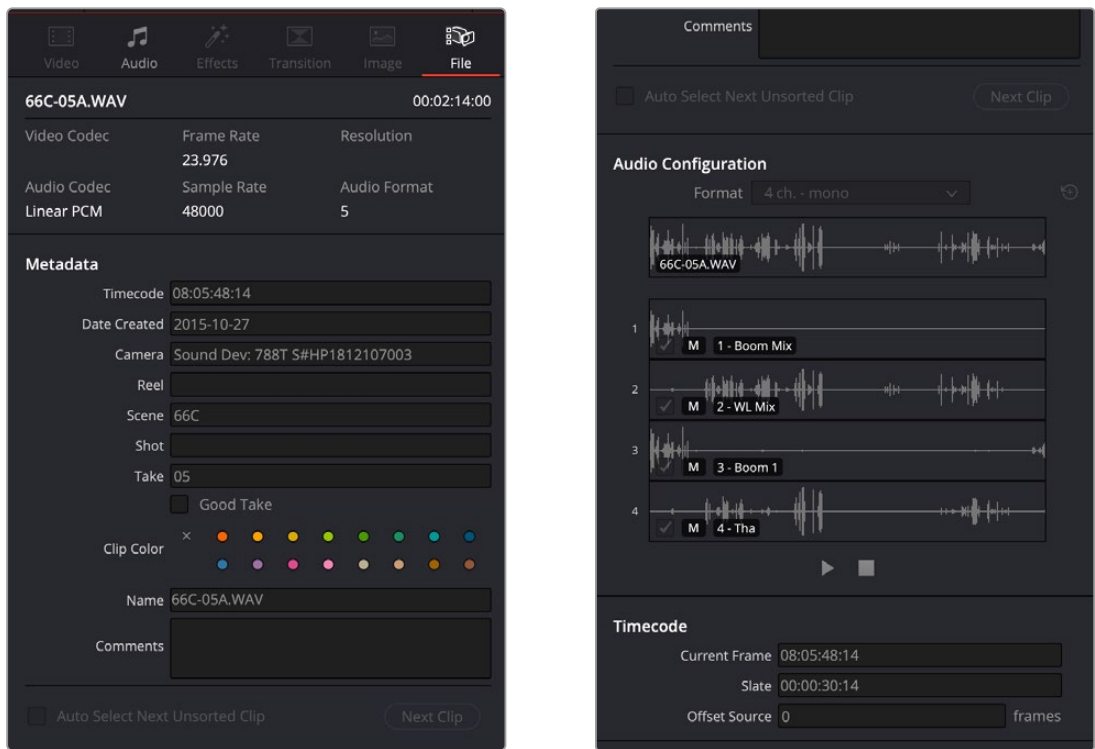


The Image Inspector controls for BRAW footage

The Image panel contains groups of parameters that correspond to every camera raw media format that's supported by DaVinci Resolve. Using these parameters in the Image panel, you can override the original camera metadata that was written at the time of recording and make simultaneous adjustments to camera raw media throughout your project.

For a detailed explanation of each of the RAW camera parameters supported by DaVinci Resolve, see *Chapter 7, "Camera Raw Settings."*

File



The File Inspector controls

Metadata

The File panel of the Inspector provides a consolidated way to view and edit a subsection of a clip's most commonly used media file metadata. It's easily accessible in the Inspector across the Media, Cut, Edit, and Fairlight pages.

The tab is composed of the following parts:

- **Clip Details:** Presents data about the clip's data format (codec, resolution, frame rate, etc.).
- **Metadata:** Presents a reduced set of common metadata fields for quick user entry.

Timecode: The start timecode of the clip. This field is editable if you want to manually change the clip's starting timecode.

Date Created: The date that the clip was created. This field is editable if you want to manually change the clip's creation date.

Camera: Sets the Camera # metadata.

Reel: Sets the Reel/Card ID.

Scene: The Scene number of the clip.

Shot: The Shot letter/number of the clip.

Take: The Take number of the clip.

Good Take: This checkbox indicates if the clip is a good or circled take.

Clip Color: Assign a specific color to a clip that is reflected in the Timeline.

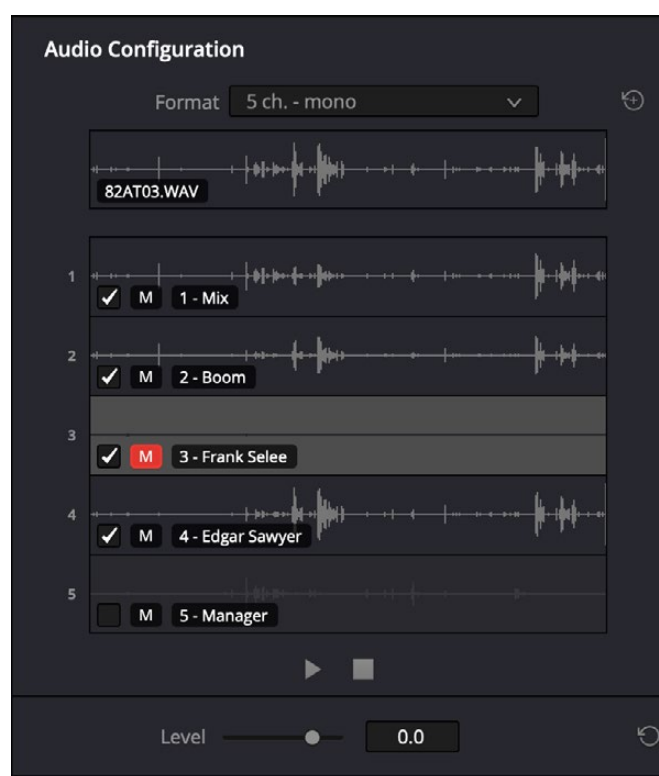
Name: This can be entered manually and changes a clip's name in that specific timeline only.

Comments: Add a text description to the clip.

Auto Select Next Unsorted Clip: When this box is checked, the next clip in the Media Pool is selected when you hit the Return button after entering a metadata field, and the cursor is automatically placed in the same field. This allows rapid sequential metadata entry without having to manually click to load each individual clip in the Media Pool. The Next Clip button will select the next clip in the Media Pool, regardless of the checkbox status.

Audio Configuration

The File tab in the Inspector now has an Audio Configuration pane that handles the controls that were formerly handled by Clip Attributes in the Media Pool (though that option is still available). The Audio Configuration pane provides a more intuitive and visual way of changing the track properties of an audio clip. Simply click on an audio clip in the Media Pool or Timeline, and then on the File Inspector to reveal this interface.



The Audio Configuration panel. Track 3 has been muted and Track 5 has been disabled.

The pane features a per-channel waveform display for all tracks within a multichannel audio file. If the tracks have been named in the audio recorder, these names will appear over their respective tracks as well. The Audio Configuration panel can preview up to 36 tracks of audio.

At the top of the pane, a composite waveform is shown of all the tracks and is updated depending on the mute status of individual tracks. By default this composite is heard when the Play button is activated, and all channels are audible.

A format menu allows you to choose common configurations for your source audio file without cumbersome manual re-routing. Custom routing can still be accommodated by choosing custom in the drop-down, which brings up the older clip attributes for situations that require less common configurations.

Audio for a selected source or timeline clip can be played or skimmed by moving the cursor along the waveform, and the specific track is solo'ed when skimming or playing. The play position or track being monitored can also be switched dynamically during playback. For example, you can start playback on Track 1, then simply click on Track 2, and the playback continues from that position. These controls let you quickly skim through and identify exactly what audio is on which track for further adjustment.

Each track has two adjustments that can be made, an Enable/Disable checkbox and a Mute button.

Enable/Disable: Enabling a track makes that track available for use in editing operations. Disabling a track removes that track from use in editing operations. For example, if you disable Track 2 of a 4-track audio file, when you drag that audio file from the Media Pool into the timeline, only 3 tracks (1, 3, and 4) will come over.

These adjustments can only be made on Media Pool clips; audio clips already in the timeline will have these options disabled.

Mute: Clicking this icon will mute the track, so it is unheard, but leave the actual track in place when used in editing operations and dragged into the timeline. Option-clicking a Mute button on a channel will allow you to solo by muting all other channels.

Underneath the track layout is a simple transport control comprised of Play and Stop buttons to start and stop audio playback.

Adjusting Trim Levels of Individual Channels in a Source File

The level of individual source clip channels in the audio inspector can be adjusted via a trim control. This allows source clip audio in one channel to be brought up in level, or brought down in level as needed.

The trimmed level is internal to the source clip in the Media Pool and will be inherited whenever that source clip configuration is used on a timeline, unless it is changed.

NOTE: The trim level is totally separate from clip gain in the timeline. Be aware that trimmed levels can clip if you move them too high.



You can adjust the trim levels of individual audio channels in a multichannel clip.

Using the Trim slider

- 1 Select one or more channels in the Audio Configuration section of the File Inspector.
- 2 Adjust the Level slider at the bottom to the desired audio level.

The waveform will adjust dynamically to the level value and will display a trimmed value in dB in the channel itself. If you have multiple lanes selected, all channels will be trimmed together. Relative levels are not stored, so if you adjust one, it will then force any others in the selection to match. When no channel lanes are selected, the Level control is grayed out.

Multiple Clip Selection and Adjustments

You can select multiple audio clips and adjust their properties in the Audio Configuration pane. For example, you can select a group of audio files and remove Track 2 from all of them at once. However, the following should be noted:

- In a multiple clip selection, only the last selected clip will appear in the track layout of the Audio Configuration pane. However, the top composite waveform will be named “Multiple Clips” to let you know that more than one clip has been selected.
- Any adjustments, like muting or enabling/disabling a track, will be applied to all the selected clips at once.

Timecode

The File tab in the Inspector now has a Timecode pane that handles the types of controls that were formerly handled by Clip Attributes in the Media Pool (though that option is still available). Here you can override the timecode details for a clip or clips in the Media Pool.

- **Current Frame:** Lets you assign a new time for the timecode at the currently viewed frame of the clip.
- **Slate:** In situations where source media comes from a shoot where a timecode slate was used during the shoot, then you can assign the slate timecode as a second timecode track that can be used for various operations, without changing the primary timecode of the clip, which may already be in use for program sync.

To set appropriate Slate timecode, select a clip in the Media Pool with a visible timecode slate, and move the playhead to a frame where the timecode in the slate is clearly readable. Then, open the Timecode panel of the Clip Attributes window, and type the timecode value you see in the image into the Slate Timecode field.

- **Offset Source:** If an entire set of clips has timecode that's merely offset, you can correct the timecode offset for as many selected clips as you like.